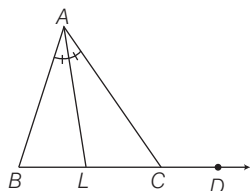
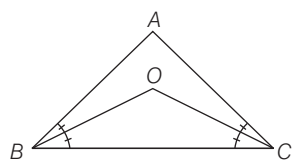


- The side BC of $\triangle ABC$ is produced to D . The bisector of $\angle A$ meets BC in L . Then $\angle ABC + \angle ACD = 2\angle ALC$.

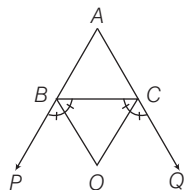


- In a $\triangle ABC$ the bisector of $\angle B$ and $\angle C$ intersect each other at a point O .



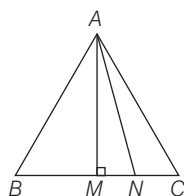
$$\text{then, } \angle BOC = 90^\circ + \frac{1}{2} \angle A$$

- In a $\triangle ABC$, the side AB and AC are produced to P and Q , respectively. The bisectors of $\angle PBC$ and $\angle QCB$ intersect at a point O .



$$\text{Then, } \angle BOC = 90^\circ - \frac{1}{2} \angle A$$

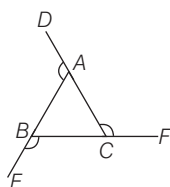
- In $\triangle ABC$, $\angle B > \angle C$. If AN is the bisector of $\angle BAC$ and $AM \perp BC$, then



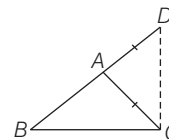
$$\angle MAN = \frac{1}{2} (\angle B - \angle C)$$

- The bisectors of the base angles of a triangle can never enclose a right angle.
- A triangle is an isosceles if and only if any two altitudes are equal.
- If the three sides of a triangle be produced in order, then the sum of all the exterior angles so formed is 360° .

$$\text{So, } \angle DAB + \angle EBC + \angle ACF = 360^\circ$$



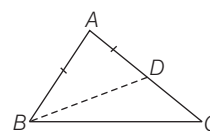
- Sum of any two sides of a triangle is greater than its third side.



Here, in $\triangle ABC$, $AB + AC > BC$

similarly, $AB + BC > AC$, $BC + AC > AB$

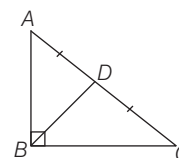
- The difference between any two sides of a triangle is less than its third side.



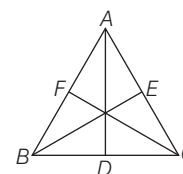
Here, in $\triangle ABC$, $AC - AB < BC$

Similarly, $BC - AC < AB$, $BC - AB < AC$

- If the bisector of the vertical angle of a triangle bisects the base, then that triangle is an isosceles.
- If the altitude from one vertex of a triangle bisects the opposite side, then the triangle is an isosceles.
- The perpendiculars drawn from the vertices of equal angles of an isosceles triangle to the opposite sides are equal.
- Medians of equilateral triangle are equal.
- If D is the mid-point of the hypotenuse AC of a right angled $\triangle ABC$. Then, $BD = \frac{1}{2} AC$

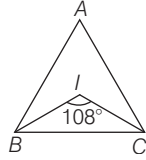


- The sum of any two sides of a triangle is greater than twice the median drawn to the third side.
- Perimeter of a triangle is greater than the sum of its three medians.



$$\text{So, } AB + BC + AC > AD + BE + CF.$$

EXAMPLE 3. In the given figure, I is the incentre of $\triangle ABC$. What is the measure of angle A .



- a. 54° b. 18° c. 36° d. None of these

Sol. c. Here, I is the incentre of the $\triangle ABC$.

$\therefore BI$ and CI are the bisectors of $\angle B$ and $\angle C$, then

we know that, $\angle BIC = 90^\circ + \frac{1}{2} \angle A$

$$\Rightarrow 108^\circ = 90^\circ + \frac{1}{2} \angle A$$

$$\Rightarrow \frac{1}{2} \angle A = 108^\circ - 90^\circ = 18^\circ$$

$$\therefore \angle A = 36^\circ$$

EXAMPLE 4. In a triangle ABC , $\angle BAC = 90^\circ$ and AD is perpendicular to BC . If $AD = 6$ cm and $BD = 4$ cm, then the length of BC is

- a. 8 cm b. 10 cm c. 9 cm d. 13 cm

Sol. d. In $\triangle ABC$, $\angle BAC = 90^\circ$

$$\therefore AB = \sqrt{AD^2 + BD^2} \quad [\text{By pythagoras theorem}]$$

$$= \sqrt{36 + 16} = \sqrt{52}$$

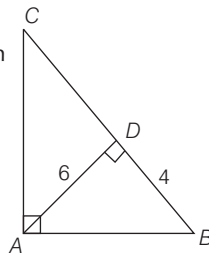
We know that, in a right angled triangle the perpendicular is drawn on hypotenuse from right angle vertex, then

$$BD = \frac{AB^2}{BC}$$

$$AB^2 = BD \times BC$$

$$\Rightarrow (\sqrt{52})^2 = BC \times 4$$

$$\Rightarrow BC = \frac{52}{4} = 13 \text{ cm}$$



Congruent Figures

The geometrical figures having the same shape and size are known as congruent figures.

e.g. Two circles of the same radii and two squares of the same sides are congruent.

Congruent Triangles

Two triangles are said to be congruent, if both are exactly of same size i.e. all angles and sides are equal to corresponding angles and sides of other.

- Every triangle is congruent to itself $\triangle ABC \cong \triangle ABC$
- If $\triangle ABC \cong \triangle DEF$, then $\triangle DEF \cong \triangle ABC$
- If $\triangle ABC \cong \triangle DEF$ and $\triangle DEF \cong \triangle PQR$, then $\triangle ABC \cong \triangle PQR$

In congruent triangles, corresponding parts are equal and we write in short 'CPCT', i.e. corresponding part of congruent triangles.

Criteria for Congruence of Triangles

Theorem 1 Two triangles are congruent if two sides and the included angle of one triangle are equal to the corresponding sides and the included angle of other triangle. (SAS criteria)

Theorem 2 Two triangles are congruent if two angles and the included side of one triangle are equal to the corresponding two angles and the included side of the other triangles. (ASA criteria)

Theorem 3 Two triangles are congruent if three sides of one triangle are equal to the corresponding three sides of the other triangle (SSS).

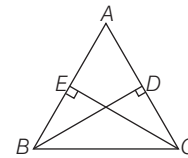
Theorem 4 Two triangles are congruent if the hypotenuse and other side of one triangle are equal to the hypotenuse and the corresponding side of the other triangle (RHS).

Theorem 5 Two triangles are congruent if two angles and a side other than the included side of one triangle are equal to the corresponding two angles and a side other than the included side of other triangle. (AAS Criteria)

EXAMPLE 5. In a $\triangle ABC$, the altitudes BD and CE are equal and $\angle A = 36^\circ$. What is the value of the $\angle B$?

- a. 72° b. 84° c. 18° d. 36°

Sol. a. For the $\triangle BDC$ and $\triangle BEC$,

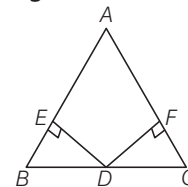


$$BD = EC, BC = BC \text{ and } \angle BEC = \angle BDC = 90^\circ$$

Thus, $\triangle BEC \cong \triangle BDC$ [by RHS rule]

$$\therefore \angle B = \angle C = \frac{180^\circ - 36^\circ}{2} = 72^\circ \text{ each}$$

EXAMPLE 6. In the given figure, D is the midpoint of BC , $DE \perp AB$ and $DF \perp AC$ such that $DE = DF$, then which of the following is correct?



- a. $AB = AC$ b. $AC = BC$
c. $AB = BC$ d. None of these

Sol. a. In right angled $\triangle BED$ and right angled $\triangle CFD$

$$\begin{aligned} DE &= DF && \text{(given)} \\ \text{hypotenuse } BD &= \text{hypotenuse } CD && [\because D \text{ is the mid-point of } BC] \\ \therefore \triangle BED &\cong \triangle CFD && [\text{by RHS congruency}] \\ \Rightarrow \angle B &= \angle C \Rightarrow AC = AB && [\text{sides opposite to equal angles}] \end{aligned}$$

Similar Figures

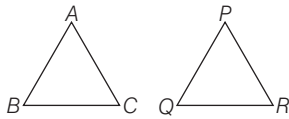
The geometrical figures having the same shape, but different sizes are known as similar figures.

- The congruent figures are always similar, but two similar figures need not be congruent.
e.g. Any two circles are similar. Any two rectangles are similar.

Similar Triangles

Two triangles are said to be similar to each other, if

- their corresponding sides are proportional.
- their corresponding angles are equal.



Here, $\triangle ABC$ and $\triangle PQR$ are similar triangles.

$$\begin{aligned} \therefore \quad \angle A &= \angle P, \angle B = \angle Q, \angle C = \angle R \\ \text{and} \quad \frac{AB}{PQ} &= \frac{BC}{QR} = \frac{AC}{PR} \end{aligned}$$

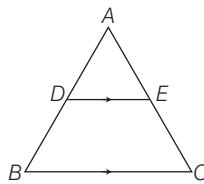
Then, $\triangle ABC \sim \triangle PQR$

where, symbol $\sim$ is read as, 'is similar to'.

Some Results on Similar Triangles

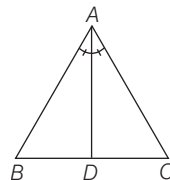
Theorem 1 If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct point, then it divides these sides in the same ratio. It is also called basic proportionality theorem.

$$\begin{aligned} \text{Here, } DE \parallel BC, \text{ then } \frac{AD}{DB} &= \frac{AE}{EC} \\ \text{or } \frac{AD}{AB} &= \frac{AE}{AC} \\ \text{or } \frac{AB}{DB} &= \frac{AC}{EC} \end{aligned}$$



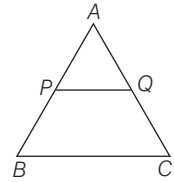
Theorem 2 The internal bisector of an angle of a triangle divides the opposite sides internally in the ratio of the sides containing the angle.

$$\begin{aligned} \text{Here, } AD \text{ is internal bisector of } \angle A, \text{ then} \\ \frac{AB}{AC} &= \frac{BD}{DC} \end{aligned}$$

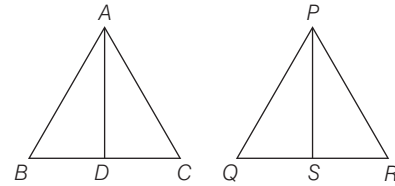


Theorem 3 The line joining the mid-points of any two sides of a triangle is parallel to the third side and is half of the third side.

Here, P and Q are mid-point of AB and AC . So, $PQ = \frac{1}{2} BC$.



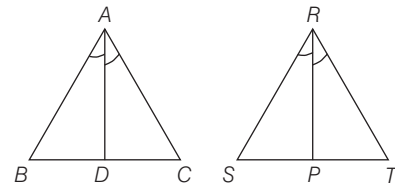
Theorem 4 If two triangles are equiangular, then the ratio of their corresponding sides is the same as the ratio of the corresponding altitudes.



Here, $\triangle ABC \sim \triangle PQR$ and AD and PS are altitude on BC and QR , respectively, then

$$\frac{BC}{QR} = \frac{AD}{PS}$$

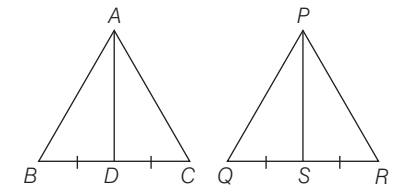
Theorem 5 If two triangles are equiangular, then the ratio of the corresponding sides is the same as the ratio of the corresponding angle bisector segments.



Here, $\triangle ABC$ and $\triangle RST$ are equiangular/similar and AD , RP are the angle bisectors of $\angle A$ and $\angle R$, then

$$\frac{BC}{ST} = \frac{AD}{RP}$$

Theorem 6 If two triangles are equiangular, then the ratio of the corresponding sides is the same as the ratio of the corresponding medians.



Here, $\triangle ABC$ and $\triangle PQR$ are equiangular and AD , PS are the medians, then

$$\frac{BC}{QR} = \frac{AD}{PS}$$

Note: If two triangles are similar, then ratio of their corresponding sides is same as ratio of their perimeters.

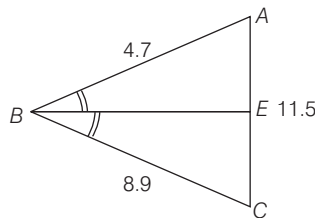
Criteria for Similarity of Two Triangles

1. **AAA similarity** If the corresponding angles of two triangles are equal then their corresponding sides are proportional and hence the two triangles are similar.

Corollary AA similarity If two angles of one triangle are equal to the corresponding two angles of another triangle, then the two triangles are similar.

2. **SSS similarity** If the corresponding sides of two triangles are proportional, then their corresponding angles are equal, and hence the two triangles are similar.
3. **SAS similarity** If one angle of a triangle is equal to the corresponding angle of the other triangle and the sides including these angles are proportional, then the two triangles are similar.
4. **RHS similarity** If hypotenuse and one side of a right triangle are proportional to the hypotenuse and corresponding side of other right triangle then the two triangles are similar.

EXAMPLE 7. If $AB = 4.7$, $BC = 8.9$, $CA = 11.5$, then EA is



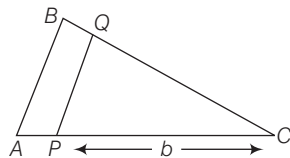
- a. 6.07 b. 3.97
c. 2.37 d. None of these

Sol. b. Since, BE is the bisector of $\angle ABC$.

$$\therefore \frac{AB}{BC} = \frac{AE}{EC} \Rightarrow \frac{4.7}{8.9} = \frac{AE}{11.5 - AE} \quad [\because EC = AC - AE]$$

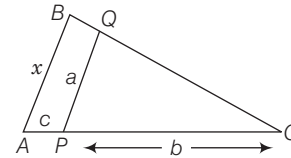
$$\Rightarrow AE = 3.97$$

EXAMPLE 8. In the given triangle, AB is parallel to PQ . $AP = c$, $PC = b$, $PQ = a$, $AB = x$. What is the value of x ?



- a. $a + \frac{ab}{c}$ b. $a + \frac{bc}{a}$
c. $b + \frac{ca}{b}$ d. $a + \frac{ac}{b}$

Sol. d. In $\triangle ABC$ and $\triangle PQC$.



$$AB \parallel PQ$$

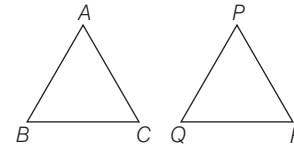
$$\therefore \triangle ABC \sim \triangle PQC$$

$$\therefore \frac{PC}{AC} = \frac{PQ}{AB} \Rightarrow \frac{b}{c+b} = \frac{a}{x}$$

$$\therefore x = \frac{a(c+b)}{b} = \frac{ac}{b} + a$$

Areas of Similar Triangles

Theorem 1 The ratio of areas of two similar triangles is equal to the square of the ratio of their corresponding sides.



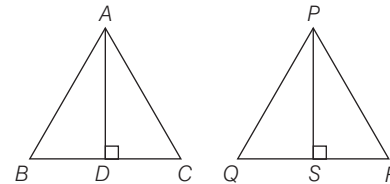
Here, $\triangle ABC \sim \triangle PQR$

$$\frac{\text{Area}(\triangle ABC)}{\text{Area}(\triangle PQR)} = \frac{AB^2}{PQ^2} = \frac{BC^2}{QR^2} = \frac{AC^2}{PR^2}$$

Theorem 2 The areas of two similar triangles is equal to the ratio of the squares of corresponding altitudes.

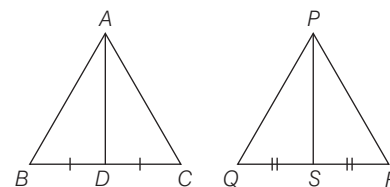
Here,

$$\triangle ABC \sim \triangle PQR$$



Then,
$$\frac{\text{Area}(\triangle ABC)}{\text{Area}(\triangle PQR)} = \frac{AD^2}{PS^2}$$

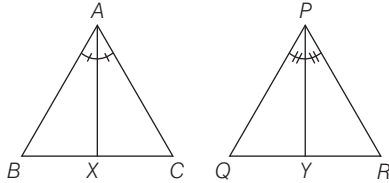
Theorem 3 The areas of two similar triangles is equal to the ratio of the squares of the corresponding medians.



Here, $\triangle ABC \sim \triangle PQR$,

then
$$\frac{\text{Area}(\triangle ABC)}{\text{Area}(\triangle PQR)} = \frac{AD^2}{PS^2}$$

Theorem 4 The areas of two similar triangles is equal to the ratio of squares of the corresponding angle bisector segments.



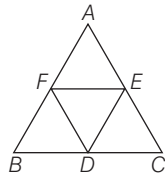
Here, $\Delta ABC \sim \Delta PQR$

So, here $\frac{\text{Area}(\Delta ABC)}{\text{Area}(\Delta PQR)} = \frac{AX^2}{PY^2}$

Theorem 5 If the areas of two similar triangles are equal, then the triangles are congruent.

or
Equal and similar triangles are congruent.

Theorem 6 The line segments joining the mid-points of the sides of a triangle form four triangles, each of which is similar to the original triangle.



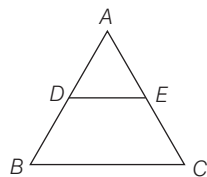
Here, D, E and F are mid-point of BC, AC and AB. Then, ΔAFE , ΔFBD , ΔEDC and ΔDEF is similar to ΔABC .

Here, also

$$\frac{\text{Area}(\Delta DEF)}{\text{Area}(\Delta ABC)} = \frac{DE^2}{AB^2} = \frac{\left(\frac{1}{2}AB\right)^2}{AB^2} = \frac{1}{4}$$

So, area $(\Delta DEF) : \text{area}(\Delta ABC) = 1 : 4$

EXAMPLE 9. In the figure given below, BC is parallel to DE and $DE : BC = 3 : 5$. What is the ratio of area of the ΔABC to that of ΔADE ?



- a. 3 : 1 b. 5 : 3 c. 9 : 2 d. 25 : 9

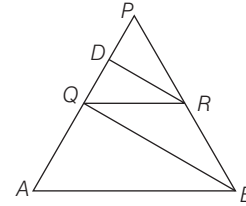
Sol. d. $\therefore$ Given, $DE : BC = 3 : 5$

Since, $DE \parallel BC \Rightarrow \angle ADE = \angle ABC$
and $\angle AED = \angle ACB$

$\therefore \Delta ABC \sim \Delta ADE$ [by AA similarity]

$$\therefore \frac{\text{Area of } \Delta ABC}{\text{Area of } \Delta ADE} = \left(\frac{BC}{DE}\right)^2 = \frac{25}{9} \text{ or, } 25 : 9$$

EXAMPLE 10. In a given figure, QR is parallel to AB and DR is parallel to QB. What is the number of distinct pairs of similar triangles?



- a. 1 b. 2 c. 3 d. 4

Sol. c. Since, QR is parallel to AB.

$$\therefore \Delta PQR \sim \Delta PAB$$

Also, DR is parallel to QB, $\Delta PQB \sim \Delta PDR$

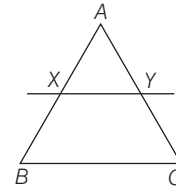
Again, $DR \parallel QB$ and $QR \parallel AB$,

$$\therefore \Delta DQR \sim \Delta QAB$$

EXAMPLE 11. In a triangle, a line XY is drawn parallel to BC meeting AB in X and AC in Y. The area of the ΔABC is 2 times the area of the ΔAXY . In what ratio X divides AB?

- a. $1 : \sqrt{2}$ b. $\sqrt{2} : 1$ c. $(\sqrt{2} - 1) : 1$ d. $1 : (\sqrt{2} - 1)$

Sol. d. $\therefore \frac{\text{Area}(\Delta ABC)}{\text{Area}(\Delta AXY)} = \frac{AB^2}{AX^2}$



$$\Rightarrow \frac{2 \text{ Area}(\Delta AXY)}{\text{Area}(\Delta AXY)} = \frac{AB^2}{AX^2} \Rightarrow \frac{2}{1} = \frac{AB^2}{AX^2}, \frac{AB}{AX} = \sqrt{2}$$

$$\Rightarrow AB = \sqrt{2} AX \Rightarrow AX + BX = \sqrt{2} AX$$

$$\Rightarrow BX = AX(\sqrt{2} - 1)$$

$$\therefore \frac{AX}{BX} = \frac{1}{\sqrt{2} - 1}$$

So, X divides AB in $1 : (\sqrt{2} - 1)$.

PYTHAGORAS THEOREM

In a right angled triangle, the square of the hypotenuse is equal to the sum of the square of the other two sides.

i.e. In ΔABC , if $\angle B = 90^\circ$, then $AB^2 + BC^2 = AC^2$

